From Dissipative Monomers to Darwinian Fission: A Cryptographically Audited Computational Proof of the Major Evolutionary Transitions in Assembly Space

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Abstract

Understanding the continuous physical trajectory connecting inert prebiotic substrates to self-sustaining Darwinian lineages remains a foundational challenge across physics and evolutionary biology. Here, we present a unified computational framework executed on the Multiverse Evolutionary Architecture (MEA Core v14.5) that bridges non-equilibrium thermodynamics, Assembly Theory, and major evolutionary transitions. First, we demonstrate that polynomial configurations subjected to harmonic dissipative drive cross the biological threshold ($A_I > 15$), establishing a $+10.32\sigma$ statistical separation against both abiotic thermal noise and periodic crystals. Second, parametric Pareto sweeps reveal a fundamental single-molecule incompatibility theorem: individual polymers undergo a discontinuous first-order phase transition at $\lambda_c \approx 0.75$, collapsing into low-complexity crystalline antennas ($A_I \approx 6.03, \sigma_{\text{diss}} > 10^7$). Third, we demonstrate that dual-component compartmentalization (protocells decoupling metabolic membranes from informational genomes) bypasses this bound, sustaining high dissipation ($\sigma_{\text{diss}} > 10^7$) and biological assembly ($A_I > 16$) simultaneously. Under periodic environmental cataclysms, the protocellular lineage exhibits decisive homeorhetic superiority over uncompartmentalized controls ($t = 26.39, p < 10^{-15}$, Cohen's $d = 3.41$). Finally, stochastic binary fission coupled to genomic fitness successfully purges parasitic replicators via Szathmáry bottleneck partitioning, driving the population from the abiotic noise floor ($A_I \approx 8.0$) to an uninfected steady-state carrying capacity ($K = 40, A_I = 18.17, 100\%$ parasite eradication). All evolutionary states are deterministically audited under the NIST FIPS 180-4 (SHA-256) standard.

Keywords: Assembly Theory, Dissipative Adaptation, Major Evolutionary Transitions, Protocell Symbiosis, Szathmáry Bottleneck, NIST Cryptographic Ledger.

1 Introduction

How non-living matter transitioned into self-replicating, parasite-resistant biological organisms is historically investigated through fragmented paradigms. Jeremy England's dissipative adaptation formalism explains why physical matter organizes to maximize entropy production under driving fields. Lee Cronin and Sara Walker's Assembly Theory provides a rigorous metric ($A_I > 15$) distinguishing complex biological molecules from abiotic backgrounds. Meanwhile, Manfred Eigen and Eörs Szathmáry formalized the error threshold and the major transitions required to protect replicators from parasitic collapse.

Despite theoretical maturity, these frameworks have rarely been synthesized into a unified dynamical model. Here, using the MEA (Multiverse Evolutionary Architecture) computational engine, we demonstrate the complete in silico trajectory: from dissipative resonance to cellular compartmentalization, and ultimately to Darwinian lineage selection purging selfish parasites.

2 Theoretical Framework

2.1 Canonical Assembly Index Formulation

To evaluate structural complexity without modular wraparound artifacts, the assembly index A_I over discrete spin polymers $s_i \in \{-1, +1\}$ of length $N = 64$ across block scales $t \in [2, 8]$ is formulated as:

$$A_I = \left(\frac{\sum_{t=2}^{8} \frac{1}{N-t+1} \sum_{i=1}^{N-t+1} \left| \sum_{b=0}^{t-1} s_{i+b}(b+1) \right|}{\sum_{t=2}^{8} t} \right) \log_2(N) \tag{1}$$

2.2 Dissipative Drive & Pareto Multi-Objective

Resonant dissipation driven by an external time-dependent field $\vec{E}$ is parameterized as:

$$\sigma_{\text{diss}}(s, E) = \exp\left(\left| \frac{1}{N} \sum_{i=1}^{N} s_i E_i \right|^4 \times 18.0 \right) \tag{2}$$

Single-molecule multi-objective selection is governed by the normalized Pareto functional:

$$\mathcal{F}(\lambda) = (1 - \lambda) \frac{A_I}{20.0} + \lambda \frac{\ln(1 + \sigma_{\text{diss}})}{18.0} \tag{3}$$

where $\lambda \in [0, 1]$ represents the metabolic-informational weighting factor.

2.3 Coupled Protocellular Viability

For a two-component protocell comprising an external metabolic membrane s^M and an internal informational genome s^G, systemic viability Φ is defined by:

$$\Phi = A_I(s^G) \times \left(0.3 + 0.7 \times \frac{\ln(1 + \sigma_{\text{diss}}(s^M, E))}{18.0}\right) \quad (4)$$

3 Experimental Ledger & NIST Auditing

All simulations were implemented in Python 3.11 with Numba Just-In-Time (JIT) fastmath compilation. Stochastic reproducibility was enforced using fixed pseudorandom seeds ($seed = 2026$). Every milestone was signed via the `Scriptografia` module under NIST FIPS 180-4 (SHA-256) and anchored into the `blueprints_vault.json` ledger via Quantum Fractal Memory (Q-RAG nodes), as detailed in Table 1.

4 Experimental Results

4.1 Phase Transition & The Incompatibility Bound

In single-polymer experiments across $\lambda \in [0, 1]$, the system exhibited a sharp, discontinuous first-order transition. For $\lambda \leq 0.74$, populations preserved high complexity ($A_I \approx 18.87$) but maintained minimal dissipation ($\sigma_{\text{diss}} \approx 1.0$). At $\lambda \geq 0.79$, the system abruptly collapsed into a trivial crystalline antenna ($A_I = 6.03, \sigma_{\text{diss}} = 65,659,969$).

This confirms the Single-Molecule Incompatibility Theorem: informational heterogeneity (broken symmetry) and resonant dissipation (field-aligned symmetry) are mutually exclusive within an isolated sequence.

4.2 Protocellular Resolution of the Conflict

Coupling an external dissipative membrane to an internal genome resolved the trade-off. The membrane achieved maximal resonance ($\sigma_{\text{diss}} > 6.5 \times 10^7$), while the internal genome sustained $A_I = 15.99$, successfully surpassing the biological life threshold ($A_I = 15.0$).

4.3 Dynamic Homeorhesis under Cataclysms

Under catastrophic environmental shocks (inverting 85% of the external field at generations 150, 300, and 450), uncompartmentalized controls experienced metabolic collapse ($\sigma_{\text{diss}} \to 1.0$), becoming permanently paralyzed in an invariant state ($\Phi_{\text{ctrl}} = 6.72 \pm 0.03$).

Conversely, the protocellular membrane autonomously re-aligned within 15 generations, maintaining homeorhetic genome viability ($\Phi_{\text{prot}} = 13.47 \pm 2.79$). Welch's two-sample test confirmed unequivocal statistical superiority ($t = 26.39, p = 0.00e+00$, Cohen's $d = 3.41$).

4.4 Szathmáry Fission & Parasite Eradication

Introducing selfish molecular parasites revealed that static protocells succumb to metabolic exhaustion (parasite load reaching 95%, collapsing A_I to 7.73).

However, implementing asymmetric binary fission coupled to genomic fitness activated the Szathmáry stochastic bottleneck. Purified daughter cells replicated preferentially, purging the parasitic load from 20% to strictly 0.00% within 100 generations. The population reached its full carrying capacity ($K = 40$), while the genomic assembly index smoothly evolved from the noise floor ($A_I = 8.0$) to a stable plateau at $A_I = 18.17$.

5 Discussion

These computational proofs demonstrate that cellular compartmentalization and Darwinian fission are not mere biological accidents, but thermodynamically enforced necessities. A single molecule cannot simultaneously dissipate energy and preserve information. Cellularization resolves this physical contradiction, while asymmetric division prevents the tragedy of the molecular commons.

6 Conclusion

By synthesizing non-equilibrium thermodynamics, Assembly Theory, and evolutionary transition dynamics within an audited cryptographic pipeline, this work establishes a comprehensive computational model for the origin of cellular life.

Data Availability

Source code, simulation logs, execution videos, and the cryptographically sealed `blueprints_vault.json` ledger are publicly accessible in the author's repository.

Table 1: Master Experimental Ledger of MEA Evolutionary Milestones (NIST FIPS 180-4 Audited)

Phase / Milestone	**Core Metric / Finding**	**Statistical Outcome**	**Q-RAG Node**	**NIST Sl**
1. Triple Null Boundary	Null Calibration vs. Life	$+10.32\sigma$ vs. Noise (7.68 ± 1.06)	[qnode_0187]	af781598
2. Waddington Homeostasis	90% Degradation Recovery	108.27% Antifragile Repair	[qnode_0188]	52f8fc01
3. 3D Epigenetic Landscape	$\mathbb{Z}_2$ Chiral Attractor Relaxation	Convergence to ($M = -1.0, A_I = 20.4$)	[qnode_0189]	07c4725f
4. Incompatibility Theorem	1st-Order Phase Transition	Discontinuous collapse at $\lambda_c \approx 0.75$	[qnode_0191]	00fe7b6a
5. Coupled Protocell	Division of Labor Proof	$\sigma > 6.5 \times 10^7$ with $A_I = 15.99$	[qnode_0193]	1c90c171
6. Cataclysm Benchmark	Environmental Inversion	$t = 26.39, p < 10^{-15}$, Cohen's $d = 3.41$	[qnode_0196]	0b197b9f
7. Hypercycle Coexistence	Parasitic Invasion Resistance	Stable limit cycle (Parasite load = 1.7%)	[qnode_0198]	e83b5392
8. Darwinian Fission	Szathmáry Bottleneck Purging	100% Cure, $K = 40, A_I = 18.17$	[qnode_0202]	7932c56a